CORE S115 Fall 2026: Lab 6

One predictor

Generate some data

```
x = c(1:100)
y1 = 10 + x*0.5 + rnorm(100, 0, 5)
y2 = 20 + x*-2 + rnorm(100, 0, 5)
y3 = 40 + rnorm(100, 0, 5)

dat= data.frame(x=x,
                y1=y1,
                y2=y2,
                y3=y3)
```

Comprehension check In the simulated data, what do you think is the relationship between:

- x and y1
- x and y2
- x and y3

Fit regression model

Here is code that fits regression models for each of the three y values.

```
fit1 = lm(y1 ~ x, data = dat)
summary(fit1)
##
## Call:
## lm(formula = y1 ~ x, data = dat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -14.7233  -3.2182   0.2581   3.1236   9.6878
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 10.19118    0.93489   10.90   <2e-16 ***
## x            0.48244    0.01607   30.02   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Residual standard error: 4.639 on 98 degrees of freedom
## Multiple R-squared:  0.9019, Adjusted R-squared:  0.9009
## F-statistic:   901 on 1 and 98 DF,  p-value: < 2.2e-16
```

```
fit2 = lm(y2 ~ x, data = dat)
summary(fit2)
##
## Call:
## lm(formula = y2 ~ x, data = dat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -11.0669  -3.5815  -0.3817   3.8044  11.6852
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 19.10877    1.01101    18.9   <2e-16 ***
## x           -1.98783    0.01738  -114.4   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.017 on 98 degrees of freedom
## Multiple R-squared:  0.9926, Adjusted R-squared:  0.9925
## F-statistic: 1.308e+04 on 1 and 98 DF,  p-value: < 2.2e-16
```

```
fit3 = lm(y3 ~ x, data = dat)
summary(fit3)
##
## Call:
## lm(formula = y3 ~ x, data = dat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -11.5276  -3.3526   0.0097   3.4201  10.3911
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 4.131e+01  9.552e-01  43.244   <2e-16 ***
## x           5.118e-04  1.642e-02   0.031    0.975
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.74 on 98 degrees of freedom
## Multiple R-squared:  9.913e-06,  Adjusted R-squared:  -0.01019
## F-statistic: 0.0009715 on 1 and 98 DF,  p-value: 0.9752
```

Comprehension check What do you notice about the coefficient values and the p-values? Does this make sense? Why is the intercept different in each case?

Two predictors

Here is code that will generate y values that are a combination of two predictors, x1 and x2.

```
x1 = c(1:100)
x2 = rep(c(0, 1), times=50)

y1 = 50 + x1*0.5 + x2*-20 + rnorm(100, 0, 5)
y2 = 50 + x1*-2 + x2*0.01 + rnorm(100, 0, 10)

dat2= data.frame(x1=x1,
                 x2=x2,
                 y1=y1,
                 y2=y2)
```

Comprehension check In the simulated data, what do you think is the relationship between:

- x1, x2 and y1
- x1, x2 and y2

Fit regression model

Here is code that fits regression models with two predictors.

```
fit4 = lm(y1 ~ x1 + x2, data=dat2)
summary(fit4)
##
## Call:
## lm(formula = y1 ~ x1 + x2, data = dat2)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -13.6122  -2.4545  -0.1663   2.8165  12.3325
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  51.87259    1.07066   48.45   <2e-16 ***
## x1            0.46215    0.01659   27.86   <2e-16 ***
## x2          -19.94303    0.95765  -20.82   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.788 on 97 degrees of freedom
## Multiple R-squared:  0.9246, Adjusted R-squared:  0.9231
## F-statistic: 595.1 on 2 and 97 DF,  p-value: < 2.2e-16
```

```
fit5 = lm(y2 ~ x1 + x2, data=dat2)
summary(fit5)
##
## Call:
```

```
## lm(formula = y2 ~ x1 + x2, data = dat2)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -31.684  -7.689   1.354   7.234  39.587
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 50.51824    2.72218  18.558   <2e-16 ***
## x1          -2.01943    0.04218 -47.882   <2e-16 ***
## x2           1.30101    2.43486   0.534    0.594
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12.17 on 97 degrees of freedom
## Multiple R-squared:  0.9594, Adjusted R-squared:  0.9586
## F-statistic:  1146 on 2 and 97 DF,  p-value: < 2.2e-16
```

Comprehension check What do you notice about the coefficient values and the p-values? Does this make sense? Specifically consider the coefficient and pvalue for x2 in fit5.

Interactions

Here is code that will generate y values that are a combination of two predictors, x1 and x2, and where the two predictors interact.

```
x1 = c(1:100)
x2 = rep(c(0, 1), times=50)

y1 = 50 + x1*0.5 + x2*-20 + x1*x2*1 + rnorm(100, 0, 5)
y2 = 50 + x1*-2 + x2*10 + x1*x2*-0.01 + rnorm(100, 0, 5)

dat2= data.frame(x1=x1,
                 x2=x2,
                 y1=y1,
                 y2=y2)
```

```
fit6 = lm(y1 ~ x1 + x2 + x1:x2, data=dat2)
summary(fit6)
##
## Call:
## lm(formula = y1 ~ x1 + x2 + x1:x2, data = dat2)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -11.2853  -3.7920   0.2474   4.0020  14.8190
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  50.61364    1.49668   33.82   <2e-16 ***
```

```
## x1            0.50795    0.02592   19.59   <2e-16 ***
## x2          -21.37996    2.13260  -10.03   <2e-16 ***
## x1:x2         0.99642    0.03666   27.18   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.291 on 96 degrees of freedom
## Multiple R-squared:  0.9793, Adjusted R-squared:  0.9787
## F-statistic:  1517 on 3 and 96 DF,  p-value: < 2.2e-16
```

```
fit7 = lm(y2 ~ x1 + x2 + x1:x2, data=dat2)
summary(fit7)
##
## Call:
## lm(formula = y2 ~ x1 + x2 + x1:x2, data = dat2)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.1503  -2.4823  -0.1821   3.6042  10.9897
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 50.94352    1.45282  35.065  < 2e-16 ***
## x1          -2.02398    0.02516 -80.429  < 2e-16 ***
## x2           9.83176    2.07010   4.749 7.13e-06 ***
## x1:x2        0.01742    0.03559   0.489    0.626
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.136 on 96 degrees of freedom
## Multiple R-squared:  0.9926, Adjusted R-squared:  0.9924
## F-statistic:  4299 on 3 and 96 DF,  p-value: < 2.2e-16
```

Factors

Categorical variable with two categories

When data is categorical R automatically assigns one of the predictors – specifically the predictor which is alphabetically first. Consider the following case.

```
x = rep(c('Instrumental', 'Noise'), times=50)
x_num = ifelse(x=='Noise', 0, 1)

y = 65 + x_num*10 + rnorm(100, 0, 5)


dat3 = data.frame(x=x,
                  y=y)
```

```
fit8 <- lm(y~x, data=dat3)
summary(fit8)
##
## Call:
## lm(formula = y ~ x, data = dat3)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -14.5889  -3.0321  -0.0245   2.8974  12.5488
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  74.9859     0.7682  97.607  < 2e-16 ***
## xNoise       -8.5672     1.0865  -7.885 4.45e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.432 on 98 degrees of freedom
## Multiple R-squared:  0.3882, Adjusted R-squared:  0.3819
## F-statistic: 62.18 on 1 and 98 DF,  p-value: 4.451e-12
```

Comprehension check Why is the coefficient negative? What does this tell you about what is being considered a "baseline"? Does this make sense?

Categorical variable with more than two categories

And if the variable has more than two categories (or "levels") then it assigns the first alphabetical as the "baseline" and computes a coefficient comparing the baseline to each of the others.

```
x = rep(c('Pop', 'Instrumental', 'Noise'), times=50)
x_num = ifelse(x=='Noise', 0, 1)

y = 65 + x_num*10 + rnorm(150, 0, 5)


dat4 = data.frame(x=x,
                  y=y)
```

```
fit9 <- lm(y~x, data=dat4)
summary(fit9)
##
## Call:
## lm(formula = y ~ x, data = dat4)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.6539  -3.5016  -0.2782   4.3622  10.7909
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept)  75.6590     0.7116 106.327   <2e-16 ***
## xNoise      -10.6896     1.0063 -10.623   <2e-16 ***
## xPop         -1.4698     1.0063  -1.461    0.146
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.032 on 147 degrees of freedom
## Multiple R-squared:  0.4743, Adjusted R-squared:  0.4671
## F-statistic: 66.31 on 2 and 147 DF,  p-value: < 2.2e-16
```

Comprehension check Why is one of the coefficients negative and one positive? What does this tell you about what is being considered the "baseline"? Does this baseline make sense?

Changing what is assigned as the baseline

The following code explicitly changes the ordering of the categories, and therefore what is considered baseline

```
dat5 = dat4 %>%
  mutate(x = factor(x, levels=c('Noise', 'Instrumental', 'Pop')))
```

```
fit10 <- lm(y~x, data=dat5)
summary(fit10)
##
## Call:
## lm(formula = y ~ x, data = dat5)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.6539  -3.5016  -0.2782   4.3622  10.7909
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    64.9694     0.7116  91.305  < 2e-16 ***
## xInstrumental  10.6896     1.0063  10.623  < 2e-16 ***
## xPop            9.2198     1.0063   9.162  4.1e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.032 on 147 degrees of freedom
## Multiple R-squared:  0.4743, Adjusted R-squared:  0.4671
## F-statistic: 66.31 on 2 and 147 DF,  p-value: < 2.2e-16
```

Comprehension check What could you do if you wanted to set Pop as the baseline?

Extra comprehension check (its ok if you don't fully get this!) You will notice that only the intercept in `fit8` matches the number we were adding. The other two intercepts were off by about 10. Why do you think this is? *Hint: look at* ***x_num*** *and what role it plays in the data generation!*